When Physics Meets Economics
The Growth and Fluctuation of Company Dynamics

Question Raised: What's the relationship between Average vs. Fluctuation

A research attempt to construct a complete mathematical model for company growth dynamics.

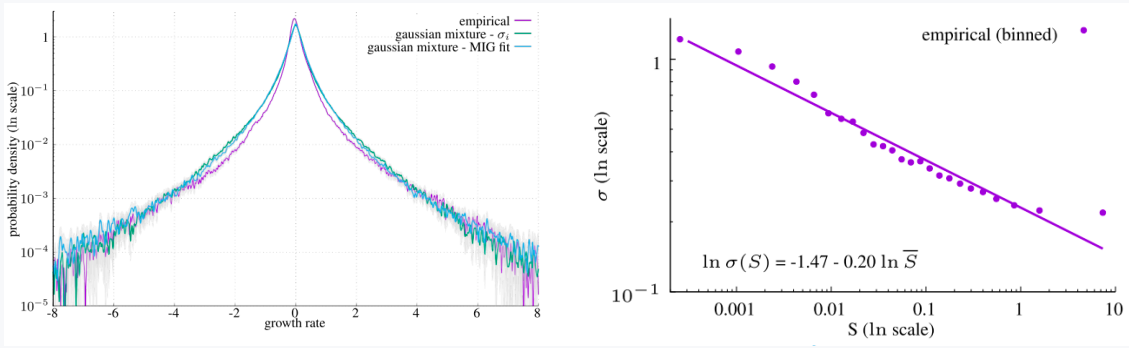
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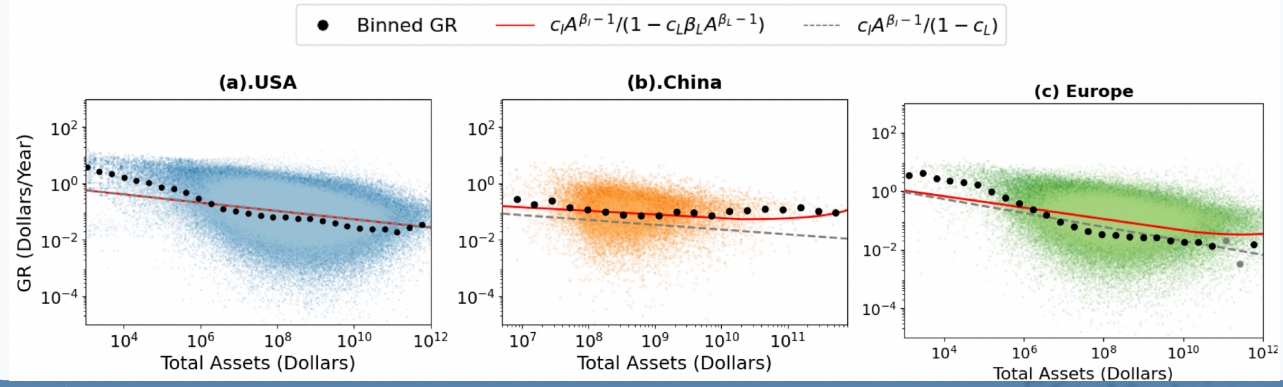
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Growth Rate Distribution P(r | s): tend-shape distribution, and $\sigma(r) \sim S^{-\alpha}$



$$r = d\ln S = \mu(S)dt + \sigma(S)dB$$

Average Growth Rate vs Size



1. From Fluctuation scaling to Growth scaling

01. Fundamental Assumptions

The growth rate distribution satisfies the scale-invariant distribution form, which can be written as:

$$P(r \mid S) = \frac{C}{\sigma(S)} \phi\left(\frac{r - \bar{r}(S)}{\sigma(S)}\right)$$

Where ϕ is a scale-independent universal function describing the scaled distribution of growth rate. C is the normalization constant.

02. Key Derivation Results

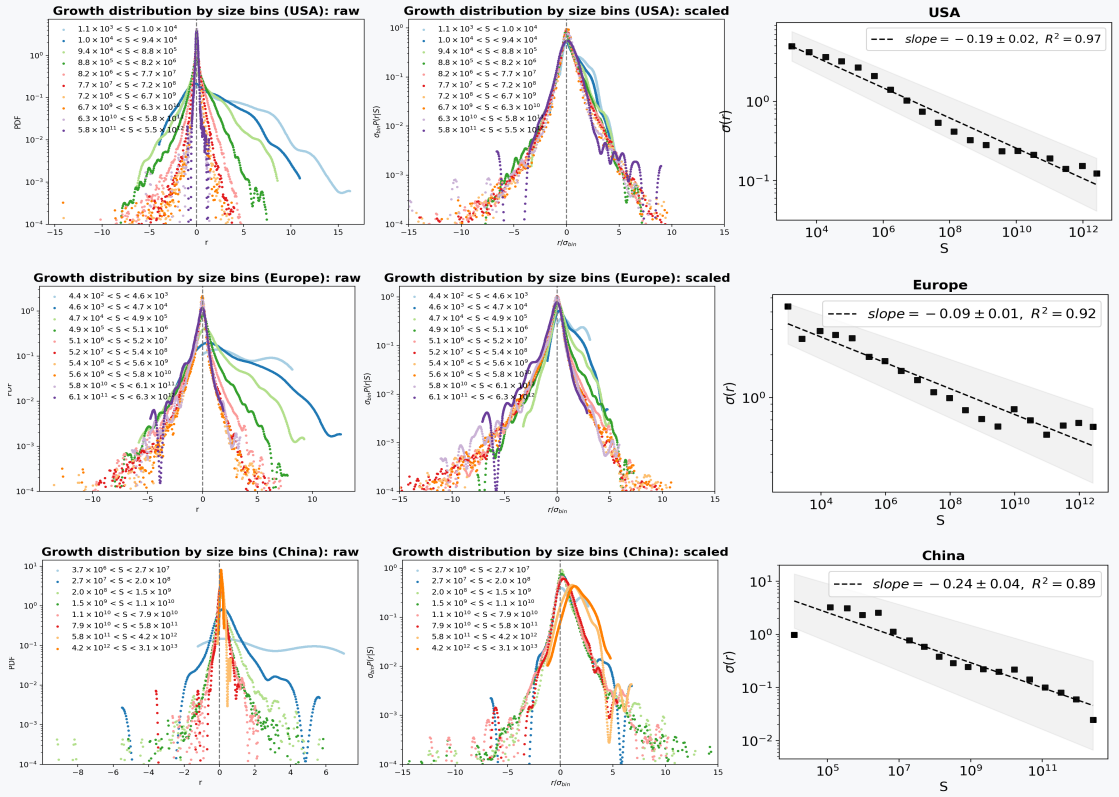
Finally, we obtain the expected logarithmic growth rate μ:

$$\mu(S) = E(r \mid S) = \int_{-\infty}^{\infty} r \frac{C}{\sigma(S)} \phi\left(\frac{r - \bar{r}(S)}{\sigma(S)}\right) dr = C\bar{r}(S) + Cm_1\sigma(S)$$

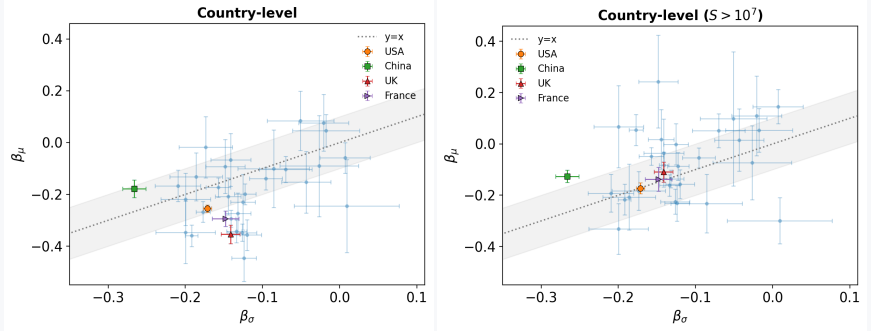
, where m_1 is the first moment of $\phi(x)$

- when $m_1 \neq 0$ (i.e., $\phi(x)$ is asymmetric) and $\bar{r}(S) \approx Const$,
  - μ and σ are linear related;
  - Furthermore, when $\mu \sim S^{-\beta_\sigma}$, μ also follows a power-law relationship with size, i.e., $\mu \sim S^{-\beta_\mu}$, and μ and σ share the **same scaling exponent,** i,e., $\beta_\mu = \beta_\sigma$

Assumptions Checks



Scaling Exponents



2. From Growth Scaling to Growth Equation

01. Average Dynamics of Company

In work [1], the mean dynamics of firm growth is known as:

$$GM(S) = \frac{dS}{Sdt} = \frac{c_I S^{\beta_I - 1}}{1 - c_L \beta_L S^{\beta_L - 1}} = \mu(S)$$

So, When $\beta_L \to 1$, we have $GM(S) \sim I/S \sim S^{\beta_I - 1}$

Note: I is net profit, β_L is the power-law exponent between assets and liabilities, and β_I is the power-law exponent between assets and net profit.,

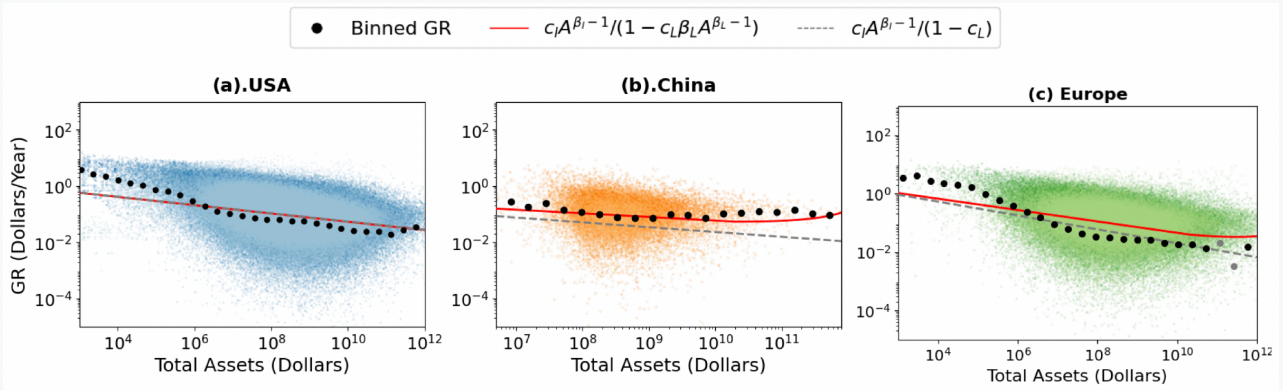
02. Key Derivation Results

Combined with the above, we know:

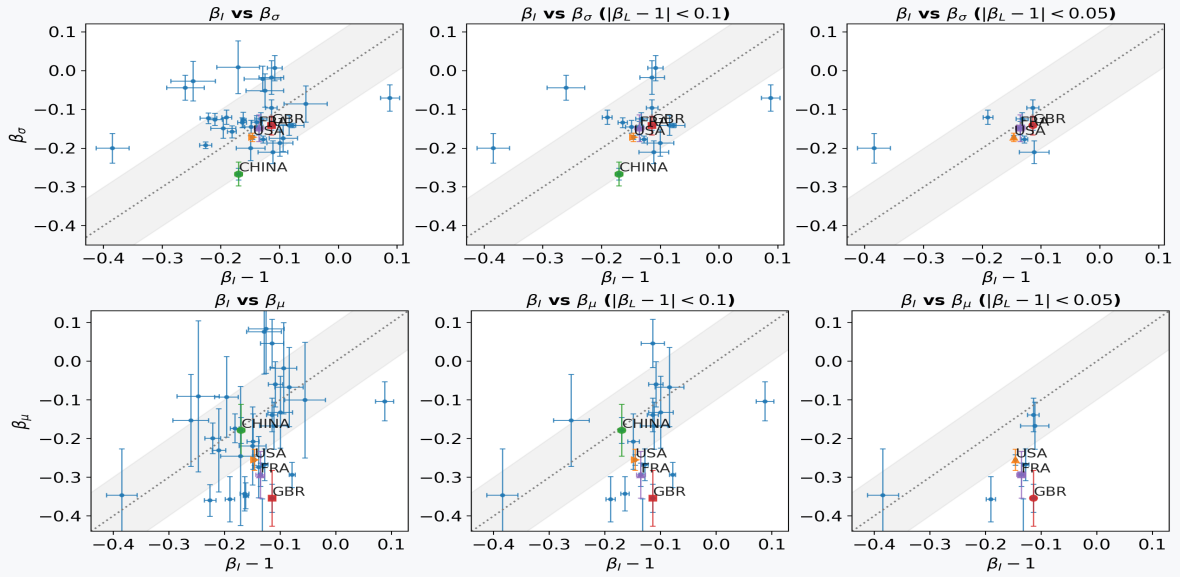
$\mu_r \sim \sigma_r \sim I/S$

When any of the three exhibits a power-law relationship with size, We could have $1 - \beta_I \sim \beta_\mu \sim \beta_\sigma$

Average Growth



Scaling Exponents



[1] **Tao R &** Zhang J, et al. Scaling laws and a general theory for the growth of public companies. arXiv preprint arXiv:2109.10379, 2024.